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Exp.NO : 10

Hamiltonian Cycles For Directed Graph ¶

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In [2]: 1 n = int(input("Enter number of vertices in the graph: "))
2 print("Enter the adjacency matrix:")
3 graph = [list(map(int, input().split())) for _ in range(n)]
4
5 def find_hamiltonian_cycles(graph):
6     n = len(graph); path = [0]; visited = [1] + [0]*(n-1); cycles = []
7     def backtrack(v):
8         if len(path) == n and graph[v][0]:
9             cycles.append(path + [0]); return
10        for u in range(n):
11            if graph[v][u] and not visited[u]:
12                visited[u] = 1; path.append(u)
13                backtrack(u)
14                path.pop(); visited[u] = 0
15        backtrack(0); return cycles
16
17 print("=====")
18 cycles = find_hamiltonian_cycles(graph)
19 if cycles:
20     print("Hamiltonian Cycles found:")
21     for cycle in cycles:
22         print(" -> ".join(map(str, cycle)))
23 else:
24     print("No Hamiltonian Cycle found")
25 print("=====")
```

Enter number of vertices in the graph: 5

Enter the adjacency matrix:

0 1 0 1 0

1 0 1 1 1

0 1 0 0 1

1 1 0 0 1

0 1 1 1 0

=====

Hamiltonian Cycles found:

0 -> 1 -> 2 -> 4 -> 3 -> 0

0 -> 3 -> 4 -> 2 -> 1 -> 0

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In []:

1

